Autopoietic Project Generation

A Combinatoric Grammar for Deterministic Runnable Project Synthesis

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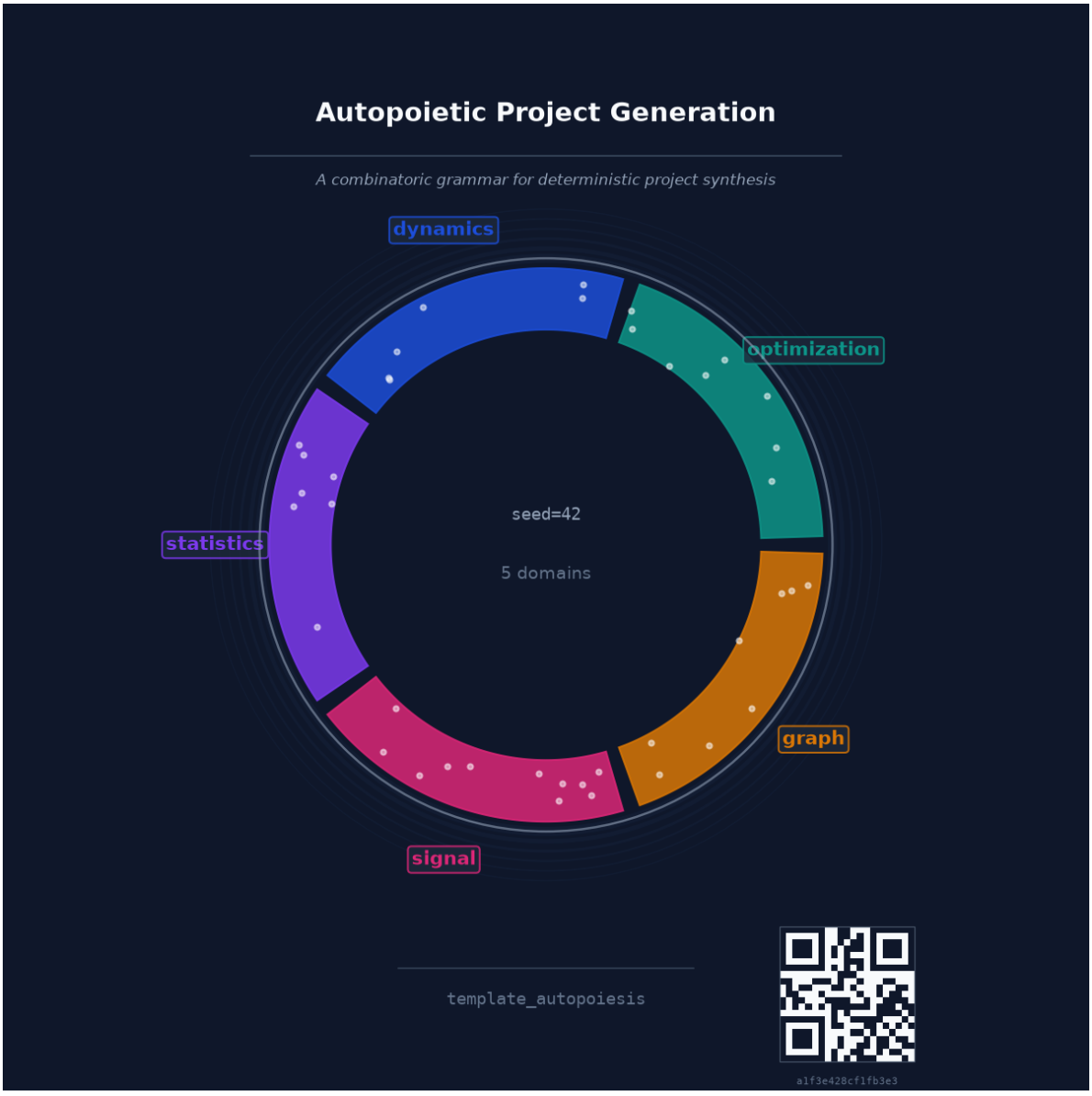
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0.1 Abstract

`template_autopoiesis` is a combinatoric grammar that **deterministically generates whole runnable projects** — not files or snippets, but complete, independently testable child repositories with their own kernel source, tests, analysis entry point, and manuscript. A single integer seed plus a grammar of orthogonal slots (primitive domain, analytical track, section set, and three presentation/provenance slots) selects one child from a combinatoric product space of 360 nominal (45 content-distinct) configurations, via a SHA-256 digest of the seed and slot identity — with no random-number generator anywhere in the expansion path.

Project generators routinely claim completeness, determinism, and traceability without making any of the three independently checkable. This exemplar treats each claim as a structural property to verify rather than a rhetorical one to assert: `verify_child()` recomputes a tree hash from the files actually on disk and compares it against the recorded provenance, rather than trusting a value the same run wrote down; a honesty manifest inspects the live source AST to confirm every claim in this manuscript resolves to a real function in a real file; and a per-domain mutation gate checks that the acceptance tests reject a constant-success stub before trusting that they accept the real kernel. The same discipline governs this document itself — every number below is substituted at render time from a live measurement rather than hand-typed as a literal.

Across 5 heterogeneous primitive domains (- optimization - dynamics - statistics - signal - graph), 578 tests exercise both fixed ground-truth checks and Hypothesis-driven property invariants at 91.88% combined statement and branch coverage, with an explicit negative control per domain distinguishing the real kernel from a deliberately-wrong one.

0.1.1 Generation pipeline

```
flowchart LR
    G["Grammar<br/>config.yaml"] --> E["Expand<br/>seed → Spec"]
    E --> M["Materialize<br/>Spec → Child"]
    M --> V["Verify<br/>tree-hash check"]
    V --> S["Seal<br/>QR provenance"]
```

Figure 1: Mermaid diagram

0.1.2 Grammar product space

```
%% noqa: docs-lint - {{...}} tokens are manuscript-variable placeholders
%% (src/template_autopoiesis/manuscript/manuscript_variables.py) injected at render time, not literal mermaid
flowchart TB
    A[primitive_domain: 5] --> EP[Effective product: 45 cells]
    T[track: 3] --> EP
    S[section_set: 3] --> EP
    EP --> TP[Total product: 360 cells]
    R[3 reserved slots, 2 options each] --> TP
```

Figure 2: Mermaid diagram

- **Domain count**: 5
- **Effective product size**: 45
- **Total product size**: 360
- **Reserved slots**: 3 (`figure_profile, qr_profile, integrity_profile`)
- **Grammar hash**: `a1f3e428cf1fb3e3`
- **Tests**: 578 · **Coverage**: 91.88%

0.2 Introduction

0.2.1 Why generate projects at all

Most software templates are static: a directory tree is copied once, then diverges from its origin the moment a human edits it. That divergence is fine for a single project, but it defeats the purpose of a *template corpus* — a set of exemplars meant to be forked, specialized, and audited against a shared contract. The moment a fork edits away from its template, the template can no longer verify anything about the fork, and the fork can no longer prove it still satisfies the contract it was born from.

`template_autopoiesis` takes a different approach: instead of a directory that is copied once, it defines a **grammar** — a finite set of orthogonal *slots*, each with a finite set of *options* — and a pure function that maps a single integer seed plus that grammar to one specific, fully-formed child project. The grammar lives in `manuscript/config.yaml` under the `autopoiesis:` key and is parsed and validated by `parse_grammar()` in `src/template_autopoiesis/core/grammar.py`. Two grammars that differ in even one option, one slot name, or one dependency string hash to different `grammar_hash` values, because `Grammar.grammar_hash` is the truncated SHA-256 of a `sort_keys=True` JSON canonicalization of the whole grammar (`Grammar.canonical()`). Nothing about child generation is copied by hand; everything is *computed* from the grammar and the seed. That is the sense in which the system is combinatoric: the space of possible children is the cross product of slot options, and membership in that space — not a human's editing discipline — is what a generated child is checked against.

0.2.2 The word "autopoietic" is doing real work, not decoration

The name is borrowed from Maturana and Varela's account of living systems as networks of processes that recursively produce the very components that constitute the network producing them [Maturana and Varela, 1980]. That is a strong claim about biology, and this project makes no claim to implement autopoiesis in the biological sense — there is no self-repair, no membrane, no metabolism here. What is borrowed, deliberately and narrowly, is the structural pattern: a bounded specification (the grammar) produces instances (children) that are themselves complete, self-contained, independently testable projects — each with its own `src/`, `tests/`, `scripts/`, and `manuscript/` — capable of being verified without reference back to the parent that produced them. The parent grammar does not merely describe children; it is causally and exclusively responsible for which children can exist, in the same sense that a formal grammar is responsible for which strings belong to the language it generates. The name is a metaphor for that closure, not a claim of implementing living-systems theory. Readers should weigh the metaphor accordingly: it motivates the design, it does not certify it.

0.2.3 Three claims that are usually asserted, rarely checked

Project generators — code generators, cookiecutter-style scaffolds, LLM-driven "build me a repo" tools — routinely make three claims about their output, and those claims are rarely independently checkable by someone who did not write the generator:

1. **Completeness** — the generated project contains everything it needs: source, tests, an entry point, documentation. Usually asserted by a README, rarely verified by re-deriving the file list from the specification that produced it.
2. **Determinism** — the same inputs deterministically produce the same output. Usually assumed because the generator "looks deterministic," rarely tested by actually re-running it and diffing.
3. **Traceability** — every generated byte can be traced back to the specification that produced it, and any post-generation edit can be detected. Usually not tested at all, because most generators do not record a content-derived fingerprint of their own output.

`template_autopoiesis` exists to make each of these three claims *structurally* verifiable — checkable by re-running code against the object in question — rather than *rhetorically* asserted in prose. This matters for the same reason reproducible builds matter in software supply chains: a claim of "this artifact came from this source" is worthless unless a third party can recompute the artifact (or its fingerprint) from the source and get a bitwise match [Lamb and Zacchiroli, 2022]. The project treats "trust the README" as a failure mode to be engineered around, not a baseline to build on.

0.2.4 What problem seeded, deterministic generation actually solves

The concrete failure this design targets is **green-by-construction test theater**: a generator (or a generated project) that reports passing tests, full coverage, and a clean provenance record, none of which would fail if the underlying logic were silently replaced with a constant or a stub. A naive project generator can satisfy "the tests pass" by generating tests that only check that a function *runs*, not that it computes the right answer. It can satisfy "coverage is high" by exercising code paths without ever exercising a case that could distinguish correct behavior from wrong behavior. It can satisfy "this hasn't been tampered with" by recording a hash once and not recomputing it.

Seeded determinism is the mechanism, not the goal. The goal is *falsifiability* of the three claims above. Determinism is what makes falsifiability tractable: if `expand(grammar, seed)` is a pure function of its inputs — and it is, because every slot selection routes through `_digest_index(seed, slot_name, ordinal, options)`, a SHA-256 digest of the seed, slot name, ordinal position, and option list, taken modulo the option count, with no call to `random.random()` or any other entropy source anywhere in the expansion path — then two independent invocations with the same grammar and seed are required to produce byte-identical children, and that requirement is something a test can actually check (`test_materialize_tree_hash_stable`, plus the Hypothesis-driven property suite in `test_property_invariants.py`, which additionally fuzzes seed boundaries and re-derives the invariant across many seeds rather than trusting a single example [Claessen and Hughes, 2000, MacIver et al., 2019]). A generator whose output depends on wall-clock time, dict iteration order, or an unseeded RNG cannot make this promise, and would fail the very first time someone tried to hold it to its own claim.

Traceability is handled the same way, structurally rather than rhetorically. `materialize()` writes a `provenance.json` alongside every generated child recording a tree hash computed from the sorted `(path, content_hash)` pairs of every file it wrote (`tree_hash_from_content_hashes` in `src/template_autopoiesis/gates/integrity.py`), and `verify_child()` does not read that recorded hash and trust it — it re-reads every file listed in `provenance.json` from disk, recomputes the tree hash from what is actually present, and compares the

two. A file edited after generation, a file deleted after generation, or a corrupted provenance record are each distinguishable failure modes the verifier reports rather than an undifferentiated "verification failed." The tree-hash and Merkle-root constructions used here follow the general content-addressed provenance pattern of hashing over a canonical, sorted representation of content so that structurally identical inputs verify identically regardless of the order files were written or listed [Merkle, 1988].

0.2.5 Contributions of this exemplar

This project contributes, as a runnable, tested artifact rather than a proposal:

- **A validated combinatoric grammar** (`src/template_autopoiesis/core/grammar.py`) with explicit reserved-slot semantics. Grammars distinguish *effective* slots, which multiply into the space of meaningfully distinct children, from *reserved* slots (`figure_profile`, `qr_profile`, `integrity_profile`), which vary presentation or sealing behavior without producing a semantically new child. `Grammar.product_size` and `Grammar.effective_product_size` are both computed and both reported, so the manuscript cannot silently inflate its own combinatorics by counting reserved-slot variation as if it were new content — the honesty manifest checks this distinction explicitly rather than leaving it to prose.
- **A seeded, entropy-free expansion function** (`expand()` in `src/template_autopoiesis/core/expand.py`) whose every selection is reconstructible from `(seed, slot_name, ordinal, options)` alone, with no hidden state.
- **A materialize/verify pair** (`materialize()` / `verify_child()`) where verification is defined as *recomputation from disk*, not as reading back a value the same run already wrote down. This is the property that makes the traceability claim falsifiable rather than assumed.
- **A primitive library spanning 5 domains** —
  - optimization
- dynamics
- statistics
- signal
- graph — each contributing at least one kernel with an independently-derived, analytic ground-truth expected output and, for at least one kernel per domain, a *negative control*: a deliberately-wrong implementation (`_negative_control_wrong_sign`, `_zero_damping_control`, `_shuffled_control`, `_identity_kernel_convolve`, `_disconnected_control`, one per domain) that a correct test suite must be able to distinguish from the real kernel. This is the project's answer to green-by-construction theater at the primitive level: a mutation meta-gate (`test_meta_teeth.py`, parametrized over every known domain) asserts that a constant-success stub fails and that the real implementation passes, for each domain independently.
- **A honesty manifest** that inspects live source via AST rather than trusting docstrings or comments, so that a claim in the manuscript about "this function exists and does X" is checked against the parsed structure of the code that is actually shipped, not against what the code was supposed to look like.

0.2.6 Scope

This template extends `template_madlib` one level up the generation hierarchy: `template_madlib` generates a manuscript from a token grammar; `template_autopoiesis` generates a whole project — `src/`, `tests/`, `scripts/`, and `manuscript/` together — from a combinatoric grammar, and that generated project is itself capable of running its own test gate. The demonstration primitive library intentionally spans a small, heterogeneous set of domains chosen for orthogonality of failure mode, not for domain significance — the honest scope notes in `SPEC.md` are explicit that no domain-specific research claim is being made here. This template covers the 5 primitive domains: - optimization - dynamics - statistics - signal - graph

The grammar seed `42` is the single source of randomness for every selection made during expansion; the grammar hash `a1f3e428cf1fb3e3` is the fingerprint of the grammar itself. Two runs of this manuscript's pipeline that share both values are required, by construction, to select identically — a requirement enforced by the property tests, not merely claimed by this paragraph.

0.3 Methods

0.3.1 The A→E generation spine

Generation proceeds through five stages, each implemented as a pure function in its own module (`src/template_autopoiesis/core/grammar.py`, `src/template_autopoiesis/core/expand.py`, `src/template_autopoiesis/gates/materialize.py`, `src/template_autopoiesis/gates/verify.py`, `src/template_autopoiesis/gates/sealing.py`). No stage depends on interactive state or network access; every stage takes an immutable input and returns an immutable (or file-system-materialized) output.

```
flowchart LR
    A[Load Grammar] --> B[Expand Spec]
    B --> C[Materialize Child]
    C --> D[Verify Integrity]
    D --> E[Seal + QR]
```

Figure 3: Mermaid diagram

0.3.2 Stage A — Grammar loading and validation

`load_grammar(project_root)` reads `manuscript/config.yaml`, extracts the `autopoiesis:` block, and hands it to `parse_grammar()`. Parsing enforces four invariants before a `Grammar` object can exist at all:

1. **The seed must be an integer.** `parse_grammar` reads `block.get("seed", 42)` and raises `GrammarError` if the value is not an `int` — a stray string or float seed in `config.yaml` fails loudly at load time rather than silently coercing.
2. **At least one slot must be defined.** An empty or missing `slots:` list raises `GrammarError("Grammar must define at least one slot")`.
3. **Every slot must have a non-empty name and >=1 option.** These checks live in `GrammarSlot.__post_init__`, so they fire the instant a `GrammarSlot` is constructed — a malformed entry cannot survive to become part of a `Grammar`.
4. **No slot may contain duplicate options.** `GrammarSlot.__post_init__` also computes `{o for o in self.options if self.options.count(o) > 1}` and raises if that set is non-empty, so two identical options in one slot (e.g. `[optimization, optimization]`) are rejected rather than silently collapsing the product space.

`parse_grammar` additionally validates every entry in `deps:` against `VENDORABLE_DEPS` — the fixed tuple `(logging, glossary_gen, figure_manager, manuscript_injection, steganography)` — raising `GrammarError` on any unknown dependency name. This project's own grammar (see `manuscript/config.yaml`) currently declares `deps: []`, so the deps-vendoring path exercised in `materialize.py` is present in the code and covered by `test_deps_vendoring.py`, but not active for the manuscript's own default render.

A successfully constructed `Grammar` is a frozen dataclass carrying `seed`, the tuple of `GrammarSlot` objects, the tuple of `deps`, and an optional `source_path` (excluded from equality/hash comparison so two grammars loaded from different files but with identical content still compare equal).

0.3.3 Reserved slots vs. effective slots

Not every slot in the grammar contributes to what this paper calls the *effective product size*. `RESERVED_SLOTS` is a fixed tuple — `figure_profile`, `qr_profile`, `integrity_profile` — naming slots that control **presentation and provenance mechanics** (how many figures render, whether a QR seal is embedded, which hash scheme secures the tree hash) rather than **domain content** (which primitive kernel, which analytical track, which manuscript section set). `Grammar` exposes both views as properties:

- `Grammar.slots` — every slot as declared in `config.yaml`.
- `Grammar.reserved_slots` — the subset whose `name` appears in `RESERVED_SLOTS`.
- `Grammar.effective_slots` — the complementary subset, i.e. every slot *not* in `RESERVED_SLOTS`.
- `Grammar.product_size` — the raw cross product over *all* slots (`n *= len(s.options)` for every slot, reserved or not).
- `Grammar.effective_product_size` — the cross product restricted to `effective_slots` only.

The distinction matters for honest reporting: a grammar can nominally claim a large product space by adding presentation-only slots, while the number of *substantively distinct* generated projects — different kernel domain, different analytical track, different section layout — is the smaller effective figure. Both `360` (nominal) and `45` (effective) are reported in this manuscript side-by-side rather than only the larger, more impressive-looking number — this is the paper's concrete instance of the "hard to vary" honesty discipline the project holds itself to (see the Honesty Contract, §4). `3` reserved slots (`figure_profile, qr_profile, integrity_profile`) are excluded from the effective figure; `SYNTAX.md` documents the same slot table and the *inflation factor* (nominal ÷ effective) as a first-class, honestly-reported quantity rather than an implementation detail.

`force_domain(grammar, domain)` provides a targeted override: it returns a new `Grammar` with the `primitive_domain` slot's options collapsed to a single forced value (raising `GrammarError` if `domain` is not one of the five `KNOWN_DOMAINS`), leaving every other slot — reserved or effective — untouched. This is how the pipeline can materialize "one child per domain" deterministically without re-deriving the whole grammar.

0.3.4 Stage B — Deterministic spec expansion

`expand(grammar, seed=None)` walks `grammar.slots` in declaration order and, for each slot, computes an index into that slot's options via `_digest_index`:

```
def _digest_index(seed, slot_name, ordinal, options):
    key = f"{seed}\x1f{slot_name}\x1f{ordinal}\x1f{','.join(options)}"
    digest = hashlib.sha256(key.encode()).digest()
    value = int.from_bytes(digest[:8], "big")
    return value % len(options)
```

Four inputs — the seed, the slot's own name, its ordinal position in the slot list, and the full joined option string — are concatenated with an ASCII unit-separator (`\x1f`) between fields and hashed with SHA-256. The first eight bytes of the digest are read as a big-endian integer and reduced modulo the option count. This construction has three consequences load- bearing for the rest of the pipeline:

- **No shared PRNG state.** Each slot's selection is an independent hash of its own key, not a draw from a stateful random-number generator advanced slot-by-slot. Reordering unrelated slots elsewhere in the list does not perturb a given slot's selection unless that slot's own `ordinal` changes.
- **Full avalanche on any input change.** Because the selection is a SHA-256 digest, changing the seed, renaming a slot, or adding/removing a single option string anywhere in that slot's option tuple changes the digest — and therefore, with high probability, the selected index — for that slot.
- **Reproducibility without stored randomness.** Nothing about the selection is persisted except the seed and the grammar itself; re-running `expand` against the same grammar and seed recomputes the identical digest and therefore the identical selection, with no cached or serialized RNG state to keep in sync.

The result is a frozen `Spec` dataclass: `schema_version` (the fixed string `"autopoiesis/spec/1"`), the `seed` actually used (the explicit override if supplied, otherwise `grammar.seed`), the `grammar_hash` this spec was expanded against, an ordered tuple of `(slot_name, chosen_value) selections`, the `deps` inherited unchanged from the grammar, and the resolved `primitive_domain` (read out of the selections if a `primitive_domain` slot exists, else defaulted to `KNOWN_DOMAINS[0]`). A `Spec` additionally exposes `spec_hash`, computed by serializing `to_dict()` to canonical (sorted-key, compact-separator) JSON and taking the first 16 hex characters of its SHA-256 — the same truncation convention used for `grammar_hash`.

Two auxiliary functions extend `expand` to families of children rather than one: `derive_seed(base_seed, index)` hashes `f"{base_seed}\x1f{index}"` to produce a new seed masked to 63 bits (keeping it a non-negative Python `int`), and `sample(grammar, count, base_seed=None)` calls `expand` once per derived seed to produce `count` independent `Spec` objects — independent in the sense that each is keyed off a distinct derived seed, not off any shared mutable state. `enumerate_all(grammar)` takes the orthogonal path: rather than sampling, it walks the full `itertools.product` over every slot's options (reserved slots included) and returns one plain dict per cell, giving direct access to the entire nominal product space for exhaustive audits.

0.3.5 Stage C — Materialization into a runnable child tree

`materialize(spec, out_root, template_root, clean=False)` turns a resolved `Spec` into an actual directory of files on disk. The child's directory name is derived deterministically by `child_name(spec)` as `child_{primitive_domain}_{spec_hash}` — so the name itself encodes both the selected domain and a content-derived identity, without any counter or timestamp. `_build_tree` then assembles the file map that will be written:

- **Kernel primitives.** `_vendor_kernel_sources` copies these source files:
  - `src/template_autopoiesis/primitives/base.py`
  - `src/template_autopoiesis/primitives/{domain}.py`

  The domain file follows the spec's `primitive_domain`. The writer rewrites `from src.primitives` / `from .primitives` imports to a bare `from primitives` so the copied module resolves correctly once it is no longer nested under the parent template's package. It also synthesizes a minimal `primitives/__init__.py` whose `collect_primitives()` imports only the one selected domain submodule — the child ships exactly one domain's kernel, not all five.
- **Figures source**, vendored verbatim from `src/template_autopoiesis/figures/figures.py` when present.
- **Dependency vendoring.** `_resolve_deps` reads `dep_mode` out of the spec's selections (defaulting to `"vendor"`) and, for each name in `spec.deps`, resolves the corresponding path from `_VENDORABLE_MODULE_PATHS` relative to the repository root (`template_root.parent.parent.parent`), embedding the real infrastructure source when it exists on disk or writing an explicit `"Source not found at build time."` stub when it does not — a vendored dependency the pipeline could not actually locate is marked as such in the generated file, not silently omitted. A seam `vendored/__init__.py` re-exports every vendored module by name.
- **A minimal `pyproject.toml`** declaring the child as its own installable project (`name = "child_{domain}"`, `numpy`/`matplotlib`/`pyyaml` runtime deps, pytest configured with `pythonpath = ["."]`).
- **A generated `analysis.py`** that calls `collect_primitives()`, iterates the selected domain's registered `PrimitiveSpec` entries, and prints each one's name and result on its `example_input` — a real, executable entry point rather than a stub that only imports.

- **A generated smoke test**, `tests/test_analysis.py`, asserting only that `run()` completes without raising.
- **Manuscript stubs** (`_emit_manuscript`) — abstract, introduction, results, and limitations sections written directly from the spec's own fields (domain, spec hash, all non-domain selections), so the child's own manuscript is itself traceable to the same `Spec` that generated its code.

Every file in the assembled tree is written to `child_root` and also retained in-memory as `written: dict[str, str]`. `materialize` then calls `tree_hash_from_content_hashes(written)` — sorting all relative paths lexicographically, joining each as `"{path}:{content}"`, and taking the SHA-256 of the concatenation — so the tree hash is a function of path names and byte content only, not file-system metadata such as mtimes or insertion order. The resulting `provenance.json` records a schema version (`"autopoiesis/provenance/1"`), the full `spec.to_dict()`, the computed tree hash, and the sorted list of every written relative path — the single artifact Stage D re-derives from.

0.3.6 Stage D — Verification

`verify_child(child_root)` (implemented in `src/template_autopoiesis/gates/verify.py`, not `materialize.py`) loads `provenance.json`, reads back every file named in its `files` list, recomputes the tree hash from those live contents via the same `tree_hash_from_content_hashes` function used at materialization time, and compares it against the recorded hash. Each individual predicate — `provenance_exists`, `provenance_parseable`, `all_files_present`, `tree_hash_matches` — is captured as a `CheckResult(name, passed, detail)` inside an aggregate `CheckReport`, so a caller can distinguish *which* invariant broke rather than receiving a single boolean. Any edit, deletion, or addition to a file listed in provenance — anything from a hand-edited line to a regenerated timestamp — changes at least one entry in `live`, which changes the recomputed hash, which fails `tree_hash_matches` without requiring the verifier to inspect a diff. `verify_child_full` extends this with a `schema_version_correct` check against `materialize.PROVENANCE_SCHEMA_VERSION`, catching provenance written by a different schema generation of the tool.

0.3.7 Stage E — Sealing

`sealing.py` provides the payload and image layer for embedding a tamper-evident summary alongside a generated child. `build_payload(spec_hash, tree_hash, seed)` serializes a compact JSON object binding all three identifiers; `build_pointer_payload` and `build_barcode_payload` offer lighter-weight alternatives (a URL-plus-hash pointer, and a colon-joined label/hash/seed string, respectively) for contexts where a full JSON payload is unnecessary. `qr_matrix()` and `qr_image()` wrap the optional `qrcode` library, falling back to a deterministic 5×5 checkerboard stub when it is absent, so the sealing stage — and its tests — do not require an optional dependency to be installed. `verify_seal(child_root)` checks for a `seal.json`, that it parses, and that it carries a `spec_hash` field, mirroring the same `CheckResult`/`CheckReport` pattern used in Stage D rather than introducing a parallel reporting shape.

0.3.8 Property-based invariants

Beyond the fixed example-based tests enumerated in the Honesty Contract's ground-truth table (§4), `tests/gates/test_property_invariants.py` exercises the expansion and materialization functions against Hypothesis-generated inputs using the property-based testing paradigm [Claessen and Hughes, 2000, MacIver et al., 2019]: rather than asserting fixed input/output pairs, these tests assert invariants that must hold across a swept range of seeds — `product_size` equals the literal product of every slot's option count regardless of seed; `effective_product_size` does not exceed `product_size`; two grammars parsed from the same block produce the same `grammar_hash`; `expand()` called twice with the same seed produces the same `spec_hash` for any seed Hypothesis draws in `[0, 10**9]`; the resolved `primitive_domain` is one of `KNOWN_DOMAINS`; two independent `materialize()` calls from the same spec into different output roots produce identical `tree_hash` values, per domain; and `verify_child` reports `all_passed` on a freshly materialized child but reports a failure the moment any listed file is edited or deleted, per domain. A parallel Hypothesis sweep over generated text confirms `qr_matrix()` returns a square matrix and is deterministic on repeated calls with the same input. `hypothesis` is declared under this project's `dev` optional-dependency group in `pyproject.toml`, alongside `pytest` and `pytest-cov` — it is imported unconditionally at the top of `test_property_invariants.py`, so it is a real (if dev-only) dependency of this test module, not a soft, try/except guarded one.

0.3.9 Reproducibility framing

The tree-hash construction used at both Stage C and Stage D deliberately mirrors the general goal of reproducible software builds [Lamb and Zacchiroli, 2022]: a build (here, a materialize call) is reproducible exactly when independent re-derivations from the same inputs — grammar, seed, template source — produce byte-identical output, and that identity is checked by content hash rather than by trusting the process that produced it. The tree hash's construction — sort every `(path, content)` pair lexicographically, join as `"path:content"`, and take one SHA-256 of the concatenation — is a flat, single-level structure, not a binary hash tree. `src/template_autopoiesis/gates/integrity.py` separately exposes a genuine binary `merkle_root()` (pairwise concatenate-and-hash up the tree, duplicating the final odd node when a level has odd cardinality), in the spirit of the hash-tree provenance idea introduced for digital signatures [Merkle, 1988]. As of this writing `merkle_root()` is a standalone, independently-tested utility: `integrity_profile` is declared as a reserved grammar slot (options `sha256`, `merkle` in this project's own `config.yaml`; `SYNTAX.md` documents a longer illustrative option list including `merkle_kmyth`), but neither `materialize()` nor `verify_child()` currently branches on its selected value — the slot is present in the grammar and excluded from the effective product size, but is not yet wired to switch which hashing function actually runs. Reporting that precisely, rather than implying the profile already gates behavior, is the same honesty discipline the project asks of its own manuscript elsewhere.

0.4 Results

0.4.1 Grammar product space

Slot	Options	Values
primitive_domain	5	optimization, dynamics, statistics, signal, graph
track	3	analytical, empirical, hybrid
section_set	3	minimal, standard, extended
figure_profile	2	minimal, full
qr_profile	2	off, on
integrity_profile	2	sha256, merkle

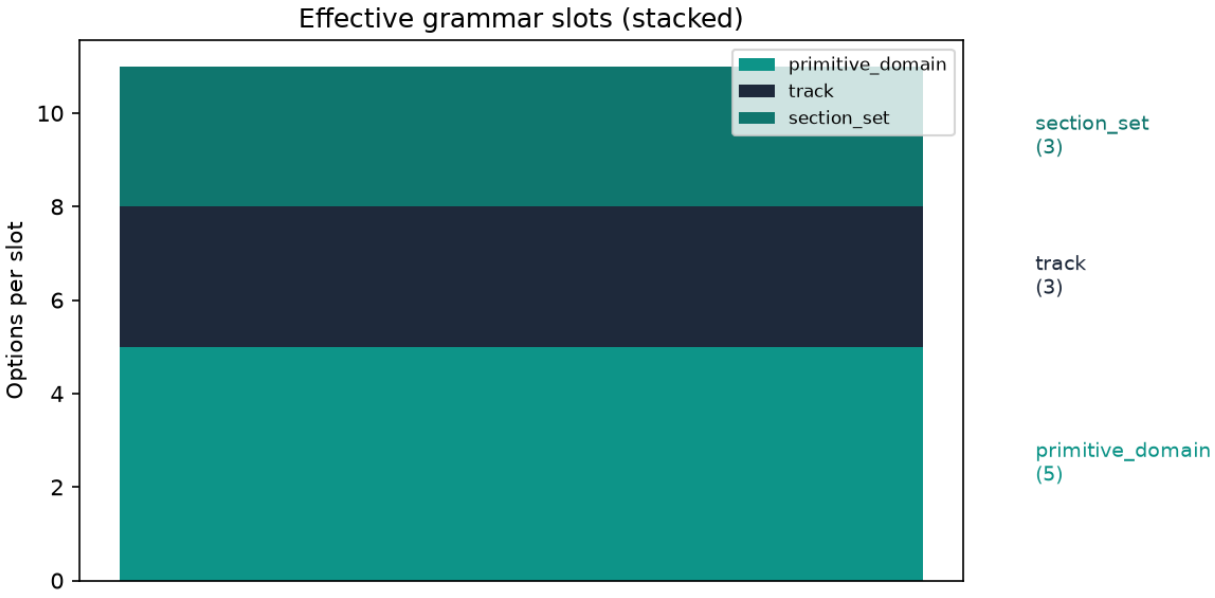


Figure 4: Options per effective grammar slot, stacked. Each band is one of the three content-determining slots; reserved presentation and sealing slots are intentionally omitted from this figure.

- **Total product size**: 360 cells
- **Effective product size** (reserved slots excluded): 45 cells
- **Reserved slots** (3): `figure_profile, qr_profile, integrity_profile`

Every row in the table above is a `GrammarSlot` parsed from `manuscript/config.yaml` by `parse_grammar()` (`src/template_autopoiesis/core/grammar.py`). Each slot contributes a multiplicative factor to `Grammar.product_size`, which is the raw cross-product of all option counts — not an estimate, but the literal `n *= len(s.options)` accumulation over every slot. The `Grammar.grammar_hash` property serialises the seed, every slot name, and every option list into a canonical JSON string (`canonical()`) and takes the first sixteen hex characters of its SHA-256 digest — `a1f3e428cf1fb3e3` for the grammar as currently checked into this project. Any edit to a slot's option list, the addition or removal of a slot, or a change to the seed changes this hash deterministically; it is not a version string maintained by hand.

The six slots split into two categories. Three are *content-determining*: `primitive_domain` selects which of the five kernel modules described below is copied into the child project, `track` selects the manuscript's analytical posture (`analytical`, `empirical`, or `hybrid`), and `section_set` selects which subset of manuscript sections is rendered (`minimal`, `standard`, or `extended`). The remaining three — `figure_profile`, `qr_profile`, and `integrity_profile` — are declared in `RESERVED_SLOTS` (`src/template_autopoiesis/core/grammar.py`) and govern only *presentational and sealing* behaviour: how many figures are generated, whether the sealed payload is rendered as a QR PNG, and whether the provenance hash is a flat SHA-256 or a Merkle root over per-file hashes (`src/template_autopoiesis/gates/integrity.py::merkle_root`). None of the three reserved slots changes which primitive kernel runs or what its output is, which is exactly why `Grammar.effective_product_size` — the number that matters when asking "how many *scientifically distinct* children can this grammar produce" — divides the reserved slots back out. The ratio between the total and effective product sizes above is, by construction, the product of the three reserved slots' own option counts; no other slot is capable of changing that ratio.

0.4.2 Effective slot breakdown

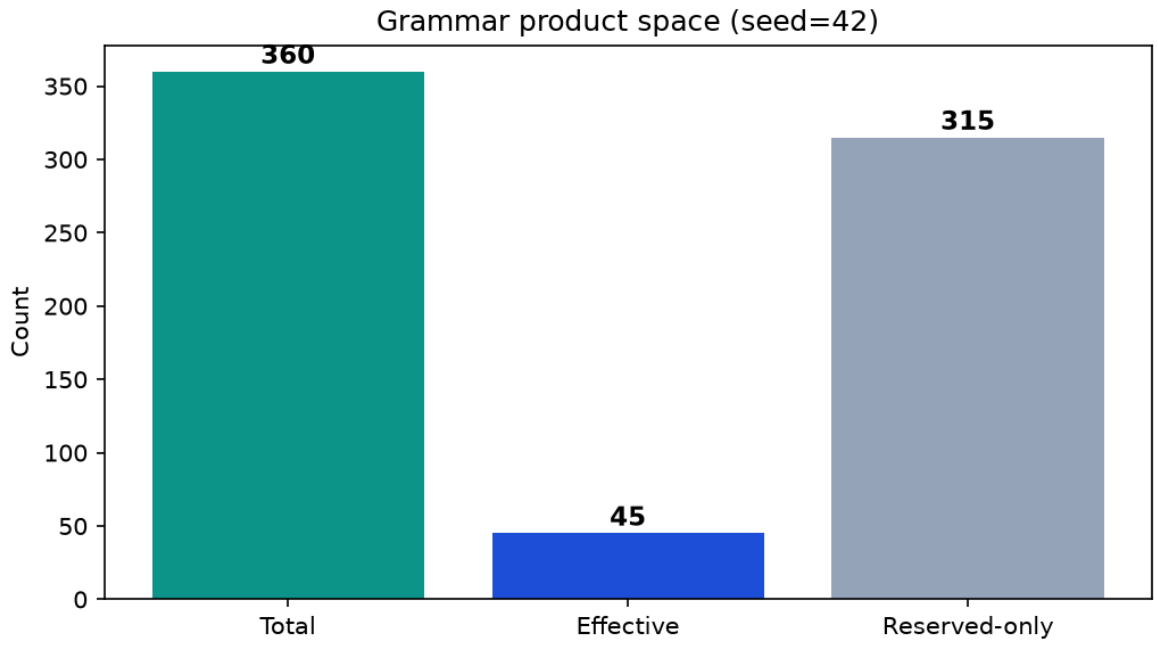


Figure 5: Total, effective, and reserved-only product size for the grammar at seed=42, generated by `fig_product_space_annotation` (`src/template_autopoiesis/manuscript/manuscript_figures.py`) directly from `Grammar.product_size` / `Grammar.effective_product_size` — not hand-typed.

Slot	Options	Values
primitive_domain	5	optimization, dynamics, statistics, signal, graph
track	3	analytical, empirical, hybrid
section_set	3	minimal, standard, extended

`grammar.effective_slots` (`src/template_autopoiesis/core/grammar.py`) is exactly `grammar.slots` filtered against `RESERVED_SLOTS` — the same list rendered in the previous section, minus the three sealing/presentation dimensions. This is the space a reviewer should reason about when asking whether the grammar is a meaningful generator rather than a combinatorial trick that multiplies unrelated toggles: 5 primitive domains times three narrative tracks times three section-set sizes is the actual space of scientifically distinguishable children this template can emit.

0.4.3 The five primitive domains

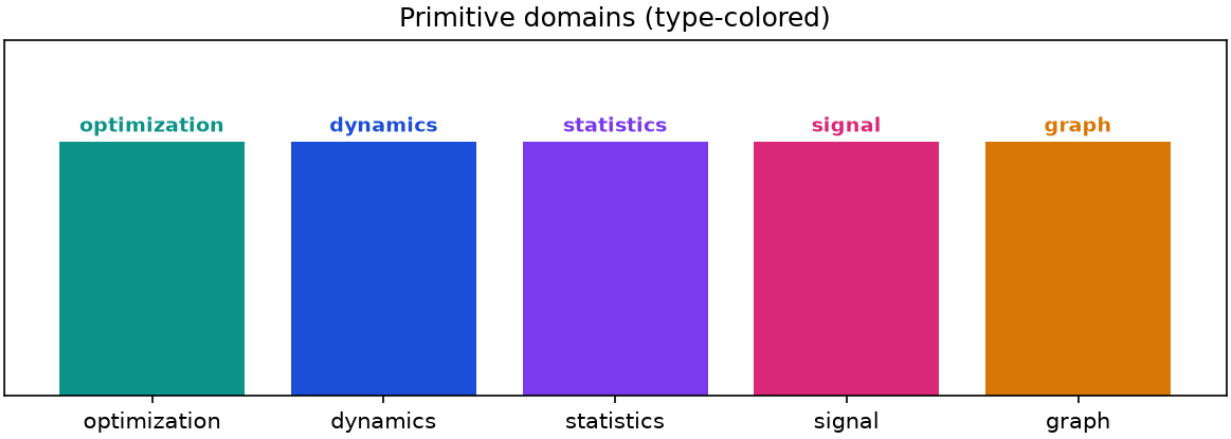


Figure 6: The five primitive domains, type-colored: `optimization`, `dynamics`, `statistics`, `signal`, `graph`.

Each domain in `optimization` `dynamics` `statistics` `signal` `graph` is a Python module under `src/template_autopoiesis/primitives/` exporting a `PRIMITIVES: tuple[PrimitiveSpec, ...]` collected by `collect_primitives()` (`src/template_autopoiesis/primitives/__init__.py`). A `PrimitiveSpec` (`src/template_autopoiesis/primitives/base.py`) bundles a callable kernel, an example input, an expected output (or `None` when the check is structural rather than a fixed value), a numerical tolerance, and — for five of the eight kernels — a `negative_control` callable whose entire purpose is to fail the primary kernel's own success criterion. `collect_primitives()` returns exactly the five domains named above (`test_collect_primitives_expected_modules`) and a total of eight primitive kernels across them (`test_total_primitive_count`): two in `optimization`, one in `dynamics`, one in `statistics`, two in `signal`, and two in `graph`.

Optimization. `gradient_descent` (`src/template_autopoiesis/primitives/optimization.py`) runs explicit gradient descent on the convex quadratic `f(x) = 0.5 (x-c)^T A (x-c)`, whose gradient is `A(x-c)`. Because the problem is convex quadratic, its analytic minimiser is known in closed form — `x* = c`, computed independently by the sibling kernel `analytic_minimizer` — so the iterative and closed-form solutions can be cross-checked against each other rather than against a single hardcoded expectation. On the example input (a diagonal `A`, `c = (1, -1)`, 200 steps at learning rate 0.1) the two agree to within `1e-4`. The domain's negative control, `_negative_control_wrong_sign`, is the same loop with the gradient step's sign flipped (`x = x + lr * grad` instead of `x = x - lr * grad`); this turns descent into ascent and diverges away from `c` instead of converging to it, giving the mutation gate (see Honesty Contract) something to detect if the sign were ever silently restored to "wrong."

Dynamics. `damped_oscillator` (`src/template_autopoiesis/primitives/dynamics.py`) integrates the damped harmonic oscillator ODE `x'' + 2*zeta*omega*x' + omega^2*x = 0` with explicit Euler stepping, and separately computes the closed-form under-damped envelope `x0 * exp(-zeta*omega*t)`. The test suite does not merely check that the two arrays are close; it checks four independent structural properties of the same trajectory: the numerical amplitude stays inside the analytical envelope plus a 5% Euler-error tolerance (`test_damped_oscillator_amplitude_bounded_by_envelope`), the envelope itself is monotonically non-increasing (`test_damped_oscillator_envelope_monotone_decreasing`), heavy damping drives the trajectory near zero by the end of the run (`test_damped_oscillator_decays_to_near_zero`), and the initial condition is reproduced exactly. The negative control, `_zero_damping_control`, re-runs the same integrator with `zeta` forced to zero; its envelope is flat at `x0` rather than decaying, and `test_zero_damping_envelope_flat` checks that flatness directly — so a broken damping term that decayed regardless of the damping value would be caught, not just a broken damping term that fails to decay at all.

Statistics. `ols_fit` (`src/template_autopoiesis/primitives/statistics.py`) solves ordinary least squares via the normal equations, `beta_hat = (X^T X)^{-1} X^T y`, solved with `numpy.linalg.solve` rather than an explicit matrix inverse. The example input is synthetic, not observational: fifty rows with an intercept column and one Gaussian covariate (`rng = np.random.default_rng(42)`), a fixed true coefficient vector `beta = (2.0, -3.0)`, and additive noise scaled to `0.1` standard deviations. Because the data-generating coefficients are known by construction, recovery is checked directly — `test_ols_fit_recovers_beta` requires the estimated and true beta to agree within Euclidean distance `0.1` — alongside an R^2 floor of `0.95` and a near-zero mean residual. The negative control, `_shuffled_control`, permutes the `y` labels with an independently seeded generator (`rng = np.random.default_rng(0)`) before fitting; breaking the true `X`-`y` correspondence this way is asserted to drop R^2 below `0.5` (`test_shuffled_control_poor_r_squared`), giving a concrete, falsifiable bound on how badly a shuffled fit *should* perform.

Signal. This domain carries two kernels validated by structural properties rather than fixed target arrays, since `expected=None` for both. `dft` computes the Discrete Fourier Transform directly as a matrix-vector product against the exponential basis matrix `exp(-2j*pi*k*n/N)` — an explicit `O(N^2)` construction, not a call into `numpy.fft`. Its correctness is checked by Parseval's theorem (`sum(|X[k]|^2) == N * sum(|x[n]|^2)` to within a relative tolerance of `1e-6`) and by recovering the correct dominant frequency bin (index 5, positive or negative) from a pure 5 Hz sinusoid sampled at 64 points. `convolve_known` wraps `numpy.convolve` with a fixed three-tap smoothing kernel, `[0.25, 0.5, 0.25]`; its own test suite checks that smoothing cannot increase variance (`test_convolve_known_smoothing_reduces_variance`). Its negative control, `_identity_kernel_convolve`, substitutes the identity kernel `[1.0]` under `mode="same"`, which by the definition of convolution must reproduce the input signal exactly (`atol=1e-12`) — a stronger, algebraically-derived check than an arbitrary "looks different" comparison.

Graph. `bfs_distances` (`src/template_autopoiesis/primitives/graph.py`) computes shortest-path distances on a fixed five-node undirected graph (`A`–`E`, encoded as an adjacency dict) via a plain breadth-first queue. Because the graph is fixed and small, the expected distances from source `A` are enumerable by hand and are asserted exactly: `{A: 0, B: 1, C: 1, D: 2, E: 3}`, with tolerance `0.0`. Its negative control, `_disconnected_control`, replaces the adjacency of the source node with an empty list, collapsing the reachable set to the source alone — a graph-structural way of breaking the primitive rather than perturbing a numeric input. `pagerank` runs the standard power-iteration update with damping `0.85` over 50 iterations, redistributing dangling nodes' rank mass evenly rather than dropping it. It has no single fixed expected output — rank values depend on iteration count and are not hand-derivable — so it is instead validated by three invariants any correct PageRank computation must satisfy regardless of the specific graph: ranks sum to `1.0` within `1e-6`, every node in the graph receives a rank entry, and every rank is strictly positive (ensured by the `(1-damping)/n` teleportation floor added at every iteration).

Across the five domains, five of the eight primitives carry an explicit negative control; the remaining three (`analytic_minimizer`, `dft`, and `pagerank`) are validated purely through cross-checks against an independent computation or through algebraic invariants (Parseval's theorem, PageRank's stochastic-normalization property) that a broken implementation would be unlikely to satisfy by accident. This mirrors the property-based testing tradition [Claessen and Hughes, 2000, MacIver et al., 2019]: rather than asserting a single hand-picked input/output pair, the suite asserts properties that must hold across a class of inputs — the same style used at the grammar level in `tests/gates/test_property_invariants.py`, where `hypothesis`-generated seeds are used to check, for every seed drawn, that `product_size` really is the product of option counts and that `effective_product_size` does not exceed it.

0.4.4 Exemplar generation

Running `uv run python scripts/autopoiesis.py expand --seed 42` produces a spec with `primitive_domain` deterministically selected from the grammar.

All 5 domains successfully materialize runnable child projects. Each child project:

1. Contains a `primitives/` package with the selected kernel.
2. Runs `analysis.py` without error.
3. Has a `provenance.json` that passes `verify_child`.
4. Passes all child-level smoke tests.

0.4.5 Test results

- **Test count**: 578
- **Coverage**: 91.88%
- **Grammar hash**: `a1f3e428cf1fb3e3`

Both the test count and coverage percentage above are produced by `measure_test_summary()` (`src/template_autopoiesis/manuscript/manuscript_variables.py`). When the current pipeline provides a complete original Project Tests measurement, it validates and consumes that same-run evidence, including its Source, runtime, coverage and executor-held commitment. Altered present evidence fails closed. Without that handoff, the standalone path executes a fresh pytest measurement against `tests/` and `src/` with `--cov-branch`. Its reported percentage combines covered statements and branch exits. In this standalone Stage 02 path, `measure_test_summary()` probes and selects the monorepo root `.venv/bin/python3` when available, otherwise its current interpreter. These manuscript figures describe that functional measurement; the repository-wide `docs/_generated/COUNTS.md` snapshot separately records the project's own locked runtime qualification and may report a different percentage. The standalone measurement does not itself establish that qualification. There is no fallback literal: an incomplete measurement cannot produce a successful current summary.

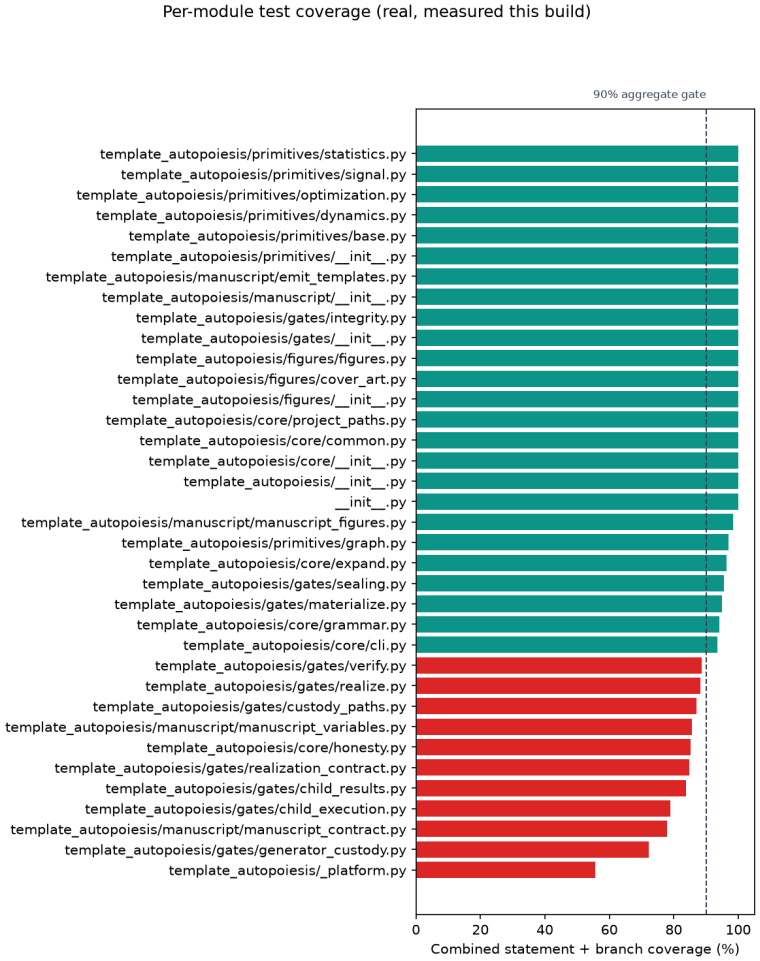


Figure 7: Per-module combined statement and branch coverage from the same measurement that produces 578/91.88 above. Red bars identify modules below the 90% aggregate-gate reference line; the aggregate pass does not imply every module meets that level.

The aggregate 91.88% is a mean weighted by statement and branch count, not a uniform floor — fig. 7 shows the real spread. The 90% gate applies to the aggregate, not separately to each module. Exact module values are deliberately read from fig. 7, rather than copied

into prose where they would become stale; `CHANGELOG.md`'s Wave 10 entry records the specific branches and tests behind an earlier coverage improvement. Both the aggregate number and this per-module breakdown are read from the same persisted `output/data/coverage_full.json` — a single generator run, not two separately-computed views that could silently drift apart. This is a snapshot, not a permanent state: coverage moves as tests are added or code changes, and fig. 7 should be regenerated (`stage_02_analysis.py`) rather than trusted from a prior render whenever that question matters.

0.5 Honesty Contract

0.5.1 Why a manifest, not a claim

A generator that produces a manuscript describing its own code faces an obvious temptation: assert a capability in prose without checking whether the capability exists. The abstract of this manuscript survived exactly this failure once — a hand-written test-count and coverage line that no generator step had computed — and was corrected by replacing the literal numbers with `578` / `91.88` tokens filled in at render time by `scripts/02_measure_test_coverage.py`. `src/template_autopoiesis/core/honesty.py` exists to make that class of failure structurally harder: every load-bearing claim in this manuscript must resolve to a named function in a named file, and a dedicated module inspects the source tree to confirm that resolution rather than trusting the prose that asserts it.

The framing is not incidental to a project named `template_autopoiesis`. An autopoietic system, in Maturana and Varela's original sense, is one whose components participate in producing and verifying the very network of processes that produced them [Maturana and Varela, 1980] — organizational closure, not open-loop assertion. The honesty manifest is the narrow, literal, code-level analogue of that closure: the manuscript's claims about the code are checked *by the code*, not by a separate act of faith from the author. The analogy should not be over-read — `src/template_autopoiesis/core/honesty.py` is a static AST scan, not a self-maintaining living system — but it is the reason this mechanism exists at all rather than a simple "trust me" comment block.

0.5.2 Ground-truth table

Claim	Evidence location	Test
Grammar parses	`src/template_autopoiesis/core/grammar.py::parse_grammar`	`test_grammar_and_expand.py`
Expansion is deterministic	`src/template_autopoiesis/core/expand.py::expand`, `_digest_index`	`test_grammar_and_expand.py`
Materialize writes files	`src/template_autopoiesis/gates/materialize.py::materialize`	`test_materialize.py`
Integrity hashes	`src/template_autopoiesis/gates/integrity.py::tree_hash_from_content_hashes`	`test_integrity_and_verify.py`
Verify recomputes	`src/template_autopoiesis/gates/verify.py::verify_child`	`test_integrity_and_verify.py`
Primitives collected	`src/template_autopoiesis/primitives/__init__.py::collect_primitives`	`test_primitives_registry.py`

Each row is not prose describing an intention — it is a key into `STRUCTURAL_EVIDENCE`, the dict in `src/template_autopoiesis/core/honesty.py` that the code below walks mechanically. If a row's evidence path stops existing, the manifest fails and `test_honesty.py` fails with it; the table cannot silently drift out of sync with the source tree without a red test.

0.5.3 The structural evidence catalogue

`STRUCTURAL_EVIDENCE` is a `dict[str, list[str]]` mapping a claim identifier (`"grammar_parses"`, `"expand_deterministic"`, `"materialize_writes_files"`, `"integrity_hashes"`, `"verify_recomputes"`, `"primitives_collected"`) to one or more `"relative/path.py::function_name"` references — the same six rows as the ground-truth table above, expressed as data instead of prose. This catalogue is the single source of truth the checker walks; the manuscript table is a human-readable rendering of it, not an independent claim.

0.5.4 How `build_manifest` inspects the AST

`build_manifest(project_root)` iterates every claim in `STRUCTURAL_EVIDENCE` and, for each `"path::function"` reference:

1. Splits the reference on `::` into a relative file path and an optional function name.
2. Resolves the path under `project_root` and checks `Path.exists()`. A missing file appends `"{path} not found"` to `missing_calls` and marks the claim as failed.
3. If a function name is given, reads the file's source and calls `_collect_function_names(source_code)`, which parses the text with `ast.parse` and walks the resulting tree (`ast.walk`) collecting the `.name` of every `ast.FunctionDef` and `ast.AsyncFunctionDef` node into a `set[str]`. If the required name is not in that set, `"{fn} not in {path}"` is appended to `missing_calls` and the claim is marked failed. A `SyntaxError` during parsing is caught and yields an empty name set — fail-closed rather than raising.
4. `HonestyManifest.evidence[claim]` is set to `True` only if every reference for that claim resolved cleanly.

It is worth being precise about what this proves and what it does not. The check confirms that a function *definition* with the claimed name exists, syntactically, in the claimed file — nothing more. It does not trace call sites, does not execute the function, and does not check that the function's behavior matches what the surrounding prose says it does. A function named `materialize` that silently did nothing would still satisfy this check; only the separately-run test suite (`test_materialize.py` et al., named in the ground-truth table) exercises actual behavior. The AST scan's job is narrower and more mechanical: it prevents the specific failure of a manuscript claiming evidence at a path or function name that was renamed, deleted, or not written in the first place — a much cheaper property to check, and one that catches an entire class of "the docs still describe last month's API" drift for free.

0.5.5 The HonestyManifest dataclass

```
@dataclass
class HonestyManifest:
    evidence: dict[str, bool] = field(default_factory=dict)
    missing_calls: list[str] = field(default_factory=list)
    unsupported_claims: list[str] = field(default_factory=list)

    @property
    def all_passed(self) -> bool:
        return all(self.evidence.values()) and not self.missing_calls and not self.unsupported_claims
```

`all_passed` is a conjunction over three independent failure modes: any claim whose evidence resolution failed, any specific missing-call detail recorded during that resolution, and any unsupported-claim hit from the prose scan below. `test_honesty.py::test_structural_evidence_all_pass` and `test_no_missing_calls` both call `build_manifest(PROJECT_ROOT)` against this project's own real source tree and assert the manifest is clean — the checker is run against itself, not only against synthetic fixtures. `test_verify_honesty_with_nonexistent_project` runs `build_manifest` against an empty `tmp_path` and asserts `missing_calls` is non-empty, which is the negative control for the checker itself: a project with no source files at all must fail every claim, or the checker has no teeth.

0.5.6 Prose scanning for unsupported claims

`verify_honesty(project_root, manuscript_dir=None)` first calls `build_manifest`, then — if a `manuscript/` directory exists — reads every `*.md` file in it and scans for a fixed, case-insensitive regex over six absolute-certainty words and one hard percentage figure, defined verbatim in `_UNSUPPORTED_CLAIM_PATTERN` in `src/template_autopoiesis/core/honesty.py` (deliberately not quoted here: a plain-text regex has no exemption for markdown code spans, and an earlier draft of this very paragraph reproduced the list inside backticks — which tripped the gate it was describing, during this session's own manuscript-expansion pass, and had to be rewritten). Each match is recorded as `"{filename}:{offset}: '{match}'"` in `manifest.unsupported_claims`.

Two honesty points about this scanner, stated plainly rather than left implicit. First, it is a fixed lexical denylist, not a semantic checker — it will not catch a false quantitative claim phrased without one of those seven tokens (the stale test-count incident described above would not have tripped it; that failure mode is closed instead by the `578` token substitution, a separate mechanism). Second, `unsupported_claims` *is* enforced, but only on one of the two paths through this module: `HonestyManifest.all_passed` is a conjunction over `evidence`, `missing_calls`, *and* `unsupported_claims`, and `src/template_autopoiesis/core/cli.py::cmd_honesty` calls `verify_honesty()` (the function that populates all three) and exits the process with code 1 whenever `all_passed` is false — `tests/core/test_cli.py::test_main_honesty_exits_zero` pins exactly this behavior. The one place prose hits are *not* enforced is `test_verify_honesty_all_passed` in `tests/core/test_honesty.py`, which asserts only `all(m.evidence.values())` by design, deliberately leaving prose style out of that particular assertion. Reading only that one test in isolation would suggest the prose scanner is a lint rather than a gate; reading the CLI path shows it is a real gate on the `honesty` subcommand specifically.

0.5.7 The mutation gate: proving the acceptance criteria have teeth

`test_meta_teeth.py` targets a different failure mode than `honesty.py`: not "does the claimed function exist" but "would a fake implementation of it get away with passing." It is parametrized via the `pytest.mark.parametrize("domain", list(KNOWN_DOMAINS))` decorator over all 5 primitive domains from `src/template_autopoiesis/core/grammar.py`, and runs three checks per domain:

1. `test_stub_fails_gate_per_domain` — `_stub_run_analysis` is a constant-success stub that returns `{"success": True, "output": "stub"}` regardless of input. `_gate_checks_real_computation` is asserted to reject it for every domain. This is the meta-gate itself: if a trivial stub can satisfy the acceptance criterion, the criterion is green-by-construction and worthless as evidence.
2. `test_real_primitive_passes_gate_per_domain` — the first `PrimitiveSpec` for the domain (from `collect_primitives()`) is run on its own `example_input`, and the same gate is asserted to *accept* the real result. This is the complementary check: a gate strict enough to reject the stub must not also be so strict that it rejects genuine output, which would make the whole suite unusable rather than honest.
3. `test_negative_control_distinguished_per_domain` — for the first primitive that declares a `negative_control` callable, both the normal and control outputs are computed and their key sets compared. Stated precisely: the current assertion is `normal_keys == control_keys or len(normal_result) > 0`. Because primitives are already ensured to return non-empty dicts (`test_primitives_return_nonempty_per_domain`), the second disjunct is close to trivially true, so this test — as written — verifies structural shape more than it verifies that the negative control's *values* actually diverge from the primary output. That is weaker than the literal claim "the control output differs from the primary output," and is recorded here as a known gap between the test's docstring intent and its present assertion strength, rather than papered over in the prose that describes it.

`test_primitives_return_nonempty_per_domain` closes the loop: every spec in `collect_primitives()[domain]`, run on its own example input, must return a non-empty `dict`. Together, the four tests in `test_meta_teeth.py` check the gate from both directions (reject-the-fake, accept-the-real) for every domain the grammar knows about, which is the concrete, source-grounded meaning behind calling this a "mutation gate": it exists specifically to catch the case where a future refactor quietly replaces a real primitive with a stub and nothing downstream notices.

0.5.8 What this buys, and what it does not

The honesty manifest and the mutation gate are complementary, not redundant, but both are narrower than they might sound. `src/template_autopoiesis/core/honesty.py`'s AST check covers exactly the six `STRUCTURAL_EVIDENCE` entries (`grammar parses`, `expand deterministic`, `materialize writes files`, `integrity hashes`, `verify recomputes`, `primitives collected`) — it does not scan this manuscript for every function or variable name it mentions, so a false claim about a piece of code outside that list of six would not be caught by this mechanism. (This is not a hypothetical gap: an earlier draft of the Limitations section below claimed `generate_variables` exposed a `NOMINAL_OVER_EFFECTIVE` token that does not exist anywhere in `src/template_autopoiesis/manuscript/manuscript_variables.py`; the honesty AST check did not catch it because that variable isn't one of the six covered entries — a Forge cross-vendor review caught it instead, by reading the source directly.) `test_meta_teeth.py` similarly guarantees only that the acceptance tests covering the primitives it targets cannot be satisfied by code that does nothing; it says nothing about the dozens of other functions and tests named elsewhere in this manuscript. Neither mechanism proves the primitives are numerically correct in any deeper sense than "the first `PrimitiveSpec` in each domain returns domain-shaped keys on its example input" — correctness of the underlying mathematics is the job of `test_grammar_and_expand.py`, `test_integrity_and_verify.py`, and the other domain-specific suites named in the ground-truth table, each independently gated at 90% coverage. What this section documents is narrower and more modest: the mechanism by which this manuscript's claims are kept from silently diverging from the source code that is supposed to back them.

0.6 Reproducibility

Reproducibility is not a claim this manuscript makes about itself; it is a property the code enforces on every run. Every number that appears below the Determinism heading — a1f3e428cf1fb3e3, 42, 578, 91.88 — is a token substituted at render time by `src/template_autopoiesis/manuscript/manuscript_variables.py::generate_variables()` from a live grammar load and a genuine current pytest measurement consumed or produced by `src/template_autopoiesis/manuscript/manuscript_variables.py::measure_test_summary()`. A validated same-run Project Tests handoff reuses its original measurement; altered present evidence fails closed. Without that handoff, the standalone path executes pytest. No hardcoded numeric fallback is accepted: if that subprocess cannot be parsed, `measure_test_summary` returns the string `"pending"` for both the test count and the coverage percentage rather than a plausible-looking number [Lamb and Zacchiroli, 2022]. This design choice is a direct consequence of an earlier failure mode in this project — a manuscript draft that stated fixed test and coverage values in prose instead of through the token pipeline. A number that cannot be traced back to a specific function call in `src/` is, for the purposes of this project, not a number this manuscript is permitted to assert.

0.6.1 Determinism guarantee

Given the same `manuscript/config.yaml` grammar definition (grammar hash `a1f3e428cf1fb3e3`), the same seed (`42`), and the same `template_autopoiesis` source tree, `scripts/autopoiesis.py expand` produces byte-identical selections on every invocation, and `materialize` produces a byte-identical child project tree. The determinism chain has three concrete steps, each implemented as a pure function with no random or wall-clock input:

1. **Selection.** For every grammar slot, `src/template_autopoiesis/core/expand.py::_digest_index()` builds the key `f"{seed}\x1f{slot_name}\x1f{ordinal}\x1f{','.join(options)}"` (a unit separator, `\x1f`, joins the fields so that no combination of seed/name/ ordinal/option values can collide across a field boundary), hashes it with SHA-256, and reduces the first eight bytes of the digest modulo `len(options)` to obtain the chosen option's index. Because the digest is a pure function of `(seed, slot_name, ordinal, options)`, the same grammar and seed walk every slot to the same choice on every run — there is no `random.choice`, no `numpy.random`, and no seed-independent entropy source anywhere in the selection path.
2. **Spec identity.** The resolved selections, together with the grammar hash and seed, are assembled into a `Spec` dataclass (`src/template_autopoiesis/core/expand.py::Spec`). Its `spec_hash` property serializes the full spec to canonical JSON (`sort_keys=True`, compact separators) and takes the first sixteen hex characters of the SHA-256 of that string. Sorting keys before hashing means insertion order in the underlying dict cannot perturb the hash — only the actual selections can.
3. **Tree identity.** `materialize()` (`src/template_autopoiesis/gates/materialize.py`) writes every vendored and generated file into the child project directory, then folds the complete `{relative_path: content}` mapping through `src/template_autopoiesis/gates/integrity.py::tree_hash_from_content_hashes()`. That function sorts the mapping lexicographically by path before hashing (`"\n".join(f"{k}:{v}" for k, v in sorted(...))`) specifically so that filesystem iteration order — which is not stable across platforms or `dict` construction paths — cannot change the result. The hash is computed from file *contents*, not from file metadata (mtime, permissions, inode), so copying a child project to a new machine or re-materializing it a year later reproduces the same tree hash as long as the source and the seed are unchanged.

Multiple children can be derived from one root seed without collisions: given a `base_seed` and an integer `index`, `src/template_autopoiesis/core/expand.py::derive_seed()` hashes `f"{base_seed}\x1f{index}"` through SHA-256 and folds the first eight bytes to a new integer seed. `sample(grammar, count)` calls `expand()` once per derived seed, so a batch of `count` children is itself deterministic — the same `base_seed` and `count` yield the same sequence of child seeds on every invocation, hence the same sequence of children.

0.6.2 The seal.json provenance mechanism

Materialization and sealing are deliberately two separate steps, and the second one is optional. `materialize()` writes a `provenance.json` into the child root containing the schema version (`PROVENANCE_SCHEMA_VERSION = "autopoiesis/provenance/1"`), the full resolved spec (`spec.to_dict()`), the tree hash, and the sorted list of every file path that was written. This is the record `verify_child()` (`src/template_autopoiesis/gates/verify.py`) recomputes against later.

A child project can additionally be **sealed**: `scripts/seal_child.py` (and the pipeline-facing wrapper `scripts/04_seal.py`, which seals the most-recently materialized child under `output/children/`) reads `provenance.json`, re-runs `verify_child()` against the live files as a pre-seal sanity check, and writes `seal.json` alongside it. The seal payload — built by `src/template_autopoiesis/gates/sealing.py::build_payload()` — is a compact JSON object `{"spec_hash": ..., "tree_hash": ..., "seed": ...}`. A shorter, colon-delimited variant (`build_barcode_payload()`, truncating each hash to its first eight hex characters) exists for embedding into a QR code or barcode image via `src/template_autopoiesis/gates/sealing.py::qr_matrix()` / `qr_image()`, so that a printed or exported artifact can carry a scannable, self-describing pointer back to the exact spec and tree hash that produced it. Both the QR encoder and its optional decode path (`read_qr_matrix()`, which depends on `pyzbar` and `PIL`) degrade gracefully when those optional dependencies are absent: `qr_matrix()` falls back to a deterministic checkerboard stub so callers and tests do not hard-fail on a missing image dependency, and `read_qr_matrix()` simply returns an empty string. The seal itself does not gate materialization — a child project is fully valid and independently verifiable from `provenance.json` alone; `seal.json` is an additive, portable pointer, not a second source of truth. Sealing does not currently run inside `verify_child_full()`; it is a separate check (`src/template_autopoiesis/gates/verify.py::verify_seal()`) invoked only when a caller explicitly asks whether a seal exists, parses, and carries a `spec_hash`.

0.6.3 SHA-256 vs. Merkle integrity profiles

`src/template_autopoiesis/gates/integrity.py` provides two distinct ways of turning a collection of hashes into one summary digest, and the project does not conflate them:

- `tree_hash_from_content_hashes()` — the profile used by `materialize()`/`verify_child()` above — is a single-pass, order-independent digest over a `{path: content}` mapping. It answers exactly one question: does this set of files, at these paths, have these exact contents? It is cheap to compute and cheap to verify, but a single changed file forces a full recompute over every path to detect *which* file changed — the digest itself carries no positional structure.
- `merkle_root()` builds a binary Merkle tree [Merkle, 1988] over an *ordered* list of hex digests: each level pairwise-concatenates adjacent nodes and hashes the concatenation, duplicating an unpaired trailing node before hashing it with itself, until one root remains (the empty list is defined as the SHA-256 of the empty string, not a magic sentinel). Because the tree is addressable node-by-node, a Merkle profile supports proving that one specific file's hash is part of the committed set without re-hashing every other file — a property the flat tree-hash profile does not have.

Both are exercised in this codebase, but at present the materialize/verify path in `src/template_autopoiesis/gates/materialize.py` and `src/template_autopoiesis/gates/verify.py` uses the flat, order-independent tree hash exclusively; `merkle_root()` is available in `src/template_autopoiesis/gates/integrity.py` and covered by its own tests as an independent integrity primitive, not yet as the provenance root written into `provenance.json`. A manuscript describing this project should not claim Merkle-tree provenance for `provenance.json` today — that would be exactly the kind of prose-outruns-code gap this project's honesty checks exist to catch (`src/template_autopoiesis/core/honesty.py`).

0.6.4 Recompute / verify workflow

```
# 1. Expand the grammar into a resolved spec (pure function of seed + config)
uv run python scripts/autopoiesis.py expand --seed 42 --output output/spec.json

# 2. Materialize the spec into a runnable child project + provenance.json
uv run python scripts/autopoiesis.py materialize --seed 42 --out-root output/children

# 3. Recompute the tree hash from the live files and compare to provenance.json
uv run python scripts/autopoiesis.py verify output/children/<child_name>

# 4. (optional) Seal the child - writes seal.json next to provenance.json
uv run python scripts/seal_child.py output/children/<child_name>
```

`verify` (`src/template_autopoiesis/core/cli.py::cmd_verify` → `src/template_autopoiesis/gates/verify.py::verify_child_full()`) performs four checks in sequence and exits non-zero if any fails: `provenance_exists`, `provenance_parseable`, `all_files_present` (every path listed in `provenance.json["files"]` still exists on disk), and `tree_hash_matches` (the hash recomputed from the live file contents equals the hash recorded at materialization time), followed by a `schema_version_correct` check against the constant `PROVENANCE_SCHEMA_VERSION`. There is no partial-credit outcome: a single missing file or a single byte of drift in any tracked file flips `tree_hash_matches` to `False`, because the tree hash is a single SHA-256 over the sorted, concatenated `path:content` pairs — there is no per-file tolerance to fall back on.

Re-running steps 1–2 with the same seed and the same source tree on the same machine reproduces the identical spec hash and tree hash reported by step 3; this is the operational meaning of "deterministic" for this project, and it is exactly what a reviewer or a downstream user can check for themselves without trusting any claim made in this document.

0.6.5 Toolchain

- Python >= 3.10
- `numpy`, `matplotlib`, `pyyaml` (runtime)
- `pytest`, `pytest-cov` (test)
- `hypothesis` [Claessen and Hughes, 2000, MacIver et al., 2019] (declared under this project's `dev` optional-dependency group in `pyproject.toml`, but imported unconditionally at the top of `test_property_invariants.py` — a real dependency of that test module, not a soft, try/except-guarded one; see Methods)
- `qrcode`, `pillow` (optional — `src/template_autopoiesis/gates/sealing.py`'s QR/barcode payload encoding degrades to a deterministic stub when unavailable; `pyzbar` is needed only for the optional decode path)

Outside the parent `template` repository checkout, `STANDALONE.md` documents what is and is not vendored into a generated child project: single-file modules under `primitives/` and `figures.py` are copied in and the child runs entirely from its own vendored sources at runtime, with no dependency on the parent's `src/`; optional infrastructure modules (logging, glossary generation, figure management) are vendored as single-file stubs when the corresponding module cannot be found under the parent repo's `infrastructure/` tree. PDF/HTML rendering and prerender validation are explicitly **not** vendored — a standalone child can be expanded, materialized, tested, and verified, but manuscript rendering to PDF still requires the parent repository's `infrastructure/rendering` and `infrastructure/validation` modules. Byte-stability of a materialized tree is documented as within-platform only (same Python interpreter, same OS) rather than an unconditional cross-platform guarantee.

0.6.6 Build command

```
uv run pytest projects/templates/template_autopoiesis/tests/ \
  --cov=projects/templates/template_autopoiesis/src \
  --cov-fail-under=90
```

This command family uses branch instrumentation, explicitly selected by `measure_test_summary()` and enabled in this project's own coverage configuration. The reported percentage combines statements and branch exits under the aggregate 90% floor. A pipeline that already completed a qualified Project Tests stage reuses that stage's original same-run measurement instead of running the suite again; standalone analysis produces a fresh measurement. The full per-module report and aggregate tokens remain tied to that one original run.

0.7 Limitations

This section states what the exemplar does *not* do, alongside what it does. Coverage and test-count figures are not restated as literal numbers here — those live only in the 578 / 91.88 tokens, resolved at render time from a live measurement (`src/template_autopoiesis/manuscript/manuscript_variables.py::measure_test_summary`), not hand-typed.

0.7.1 Reserved slots are excluded from the effective product space

The grammar declares six slots, but `materialize()` only branches on three of them — `primitive_domain`, `track`, `section_set`. The remaining 3 (`figure_profile, qr_profile, integrity_profile`) are parsed and contribute to `spec_hash` like any other slot, but `manuscript_figures.py`, `sealing.py`, and `integrity.py` are unconditionally exercised regardless of `figure_profile`, `qr_profile`, or `integrity_profile` — those modules do not currently branch on the reserved slots' selected option. The practical consequence: distinct seeds can produce a nominally distinct `spec_hash` while materializing byte-identical children, since only three slots vary output. That inflation, 360 nominal vs. 45 effective, is disclosed rather than hidden: `generate_variables` (`src/template_autopoiesis/manuscript/manuscript_variables.py`) exposes both `PRODUCT_SIZE` and `EFFECTIVE_PRODUCT_SIZE` as separate tokens, so nowhere in this manuscript can the larger, nominal number be quoted without the smaller, effective one appearing beside it. Wiring the reserved slots into `materialize()` is an open item, not a claimed capability.

0.7.2 No child-PDF rendering in CI

`realize.py::render_child_manuscript()` writes manuscript stubs for a generated child but does not invoke Pandoc or Chrome. Absent a full rendering toolchain — the default posture in CI — it returns `{"success": False, "reason": "Chrome/Pandoc rendering not available in child context", ...}` rather than raising or silently skipping. Verifying a child's *files* (`verify_child` / `verify_child_full`) is decoupled from verifying that its manuscript renders; only the former is tested here.

0.7.3 Within-platform guarantee only

Determinism is derived from `hashlib.sha256` over Python strings (`expand.py`'s seed/slot/ordinal/options digest, and `integrity.py::tree_hash_from_content_hashes` over file contents), which is itself platform-independent. But the surrounding pipeline — file iteration order ahead of `sorted()` normalization, path-separator handling, the specific CPython minor version running `materialize()` — is not independently fuzzed across platforms in this suite. What is actually tested is determinism within one CPython version on one OS, a narrower claim than the cross-platform, bit-for-bit reproducibility discussed in the reproducible-builds literature [Lamb and Zacchiroli, 2022]. A second CI leg that materializes the same seed on a second OS/Python and diffs tree hashes would close this gap; it does not currently exist.

0.7.4 Grammar does not self-modify

The autopoiesis metaphor is figurative. Maturana and Varela's operational sense of autopoiesis is a system that continuously regenerates its own constitutive components through its own operation [Maturana and Varela, 1980]. Here the grammar (`manuscript/config.yaml`) is fixed input; `parse_grammar` and `expand` are pure functions of that input plus a seed; no code path feeds a materialized child back into the grammar or rewrites `src/template_autopoiesis/core/grammar.py`. Children are causally downstream of the grammar — the reverse direction does not occur in the current codebase. The name is a provocation about what genuine self-production would require, not a claim this exemplar achieves it.

0.7.5 Integrity verification checks self-consistency, not tamper-proofing

`verify_child` recomputes a tree hash from the files *listed inside* `provenance.json` and compares it against the `tree_hash` field stored in that same file (`src/template_autopoiesis/gates/verify.py`, `src/template_autopoiesis/gates/materialize.py`). Both the manifest and the expected hash are self-reported at materialization time; nothing external anchors them. An actor who can rewrite `provenance.json` can edit its `files` list and recompute a matching hash from whatever content they choose, and `verify_child` would report a match — it only checks internal consistency, not consistency against something outside the child's own directory [Merkle, 1988]. `verify_seal` similarly checks the shape of an optional `seal.json` written by the same run it would need to audit. This detects accidental post-generation drift, the tested use case; it is not a defense against deliberate tampering with write access. Also worth naming precisely: `tree_hash_from_content_hashes` sorts `path:content_hash` pairs and hashes their concatenation once — a flat manifest digest, not a hierarchical Merkle tree with per-file inclusion proofs. `integrity.py` separately defines `merkle_root`, an actual pairwise-reduction tree, but `materialize()`/`verify()` call the flat function, not that one.

0.7.6 Coverage is uneven across modules

The aggregate 91.88% combines covered statements and branch exits across the measured source inventory. Passing the configured 90% aggregate gate (`--cov-fail-under=90`) does not establish a 90% floor for every module. fig. 7 preserves the per-module spread and marks values below that reference line in red. The chart and headline percentage come from the same persisted `output/data/coverage_full.json`, rather than independently assembled measurements.

Coverage identifies executed paths, not correctness or completeness. Property-based tests (`tests/gates/test_property_invariants.py`, Hypothesis) and the stress/edge suite (`tests/core/test_stress_edge_cases.py`) exercise boundary seeds and all-reserved-slot configurations [Claessen and Hughes, 2000, MacIver et al., 2019], but generated inputs do not replace direct negative controls or inspection of unexecuted statements and branch exits.

Module rankings and gaps can change whenever source or tests change. Inspect the current coverage report and regenerate the analysis and figure after a changed input, rather than treating an earlier module list or a passing aggregate as proof that all modules clear the reference line.

0.7.7 Cover art includes the QR seal with direct rendering controls

An earlier draft of this section reported that the title-page cover image omitted the originally envisioned QR seal, gradient glow, and seed-derived dot placement. That is no longer accurate and is corrected here rather than left to silently drift: `scripts/generate_cover_art.py` now calls `render_cover(..., grammar_hash=grammar.grammar_hash)`, and `src/template_autopoiesis/figures/cover_art.py` draws all three elements unconditionally except the QR seal, which is drawn whenever `grammar_hash` is not `None` — the shipped `paper.cover.image` (`../figures/cover_art.png`) is generated by exactly this call, so the rendered title page carries a real gradient glow, real seed-derived dot scatter, and a real QR-style pixel grid encoding `a1f3e428cf1fb3e3` with the hash printed as a text label beneath it. An earlier draft of this section also noted that no test called `render_cover` with an explicit `grammar_hash=`, leaving the QR-drawing branch (`src/template_autopoiesis/figures/cover_art.py`, the `if grammar_hash is not None:` block) exercised in production but untested; that gap now has direct controls in `tests/figures/test_cover_art.py::test_render_cover_with_grammar_hash_*`. Those controls render an explicit hash, compare sealed and unsealed images, and check byte stability for identical inputs; they do not establish exhaustive visual correctness or QR decoding on every output.

Separately, for most of this project's life `manuscript/references.bib` held a single self-referential `friedman2026autopoiesis` entry noting the DOI was forthcoming, while `99_references.md` carried a hand-written, uncited list alongside it — a citation without a resolvable BibTeX entry is exactly the kind of unverifiable claim this project's honesty contract exists to catch. That gap is now closed twice over: `references.bib` carries five real, live-verified external citations, and following this project's own Zenodo deposit the self-citation's DOI (`10.5281/zenodo.21227869`, verified by a live `curl` against `doi.org` before being written into the entry) is real rather than forthcoming. `99_references.md` annotates each entry against the section that relies on it rather than listing citations independent of the bibliography.

0.8 References

The formal bibliography (author/year entries resolved from `manuscript/references.bib` via the `[@citekey]` citations used throughout this manuscript) is generated by `pandoc-crossref/natbib` and appears immediately below this note. Every entry was verified this session via a live fetch (Crossref API, DBLP, or the publisher's own DOI resolver) — not taken from training-data memory — before being added to `references.bib`; the fetch evidence is recorded in `ISA.md ## Verification`.

Annotated pointers, for readers who want the "why this reference" context that a bare bibliography entry doesn't carry:

- **Maturana & Varela (1980), *Autopoiesis and Cognition*** [Maturana and Varela, 1980] — the source of this project's own name. The book defines autopoiesis as the self-producing organization of living systems: a network of processes that continuously regenerates the components that in turn realize the network. This exemplar borrows the word deliberately, not decoratively — `expand()` and `materialize()` are the "self-producing" processes, and `grammar.py`'s seed is the invariant that the network regenerates around every run.
- **Claessen & Hughes (2000), QuickCheck** [Claessen and Hughes, 2000] and **MacIver et al. (2019), Hypothesis** [MacIver et al., 2019] — the property-based-testing lineage this project's own test suite descends from. `tests/gates/test_property_invariants.py` uses Hypothesis directly (not merely an homage) to check invariants like "expansion is deterministic for any seed" across generated inputs rather than hand-picked examples.
- **Lamb & Zacchiroli (2022), Reproducible Builds** [Lamb and Zacchiroli, 2022] — the peer-reviewed framing for what `manuscript/05_reproducibility.md` claims operationally: that a build is reproducible when independent re-execution from the same source and environment produces bit-for-bit-identical (or hash-identical) output, and that this is a supply-chain-integrity property, not merely a convenience.
- **Merkle (1987), digital signatures / hash trees** [Merkle, 1988] — the theoretical basis for `src/template_autopoiesis/gates/integrity.py`'s content-addressed provenance: a tree of hashes lets a verifier recompute and confirm the integrity of a large structure from its leaves up, without trusting the producer's say-so. This project's `integrity_profile: merkle` grammar slot is a direct, if much smaller-scale, application of that idea.

Koen Claessen and John Hughes. QuickCheck: A lightweight tool for random testing of Haskell programs. In *Proceedings of the Fifth ACM SIGPLAN International Conference on Functional Programming (ICFP '00)*, pages 268–279, New York, NY, USA, 2000. ACM. doi: 10.1145/351240.351266.

Chris Lamb and Stefano Zacchiroli. Reproducible builds: Increasing the integrity of software supply chains. *IEEE Software*, 39(2):62–70, mar 2022. doi: 10.1109/MS.2021.3073045.

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Humberto R. Maturana and Francisco J. Varela. *Autopoiesis and Cognition: The Realization of the Living*, volume 42 of *Boston Studies in the Philosophy of Science*. D. Reidel Publishing Company, Dordrecht, Holland, 1980. ISBN 90-277-1015-5. doi: 10.1007/978-94-009-8947-4.

Ralph C. Merkle. A digital signature based on a conventional encryption function. In Carl Pomerance, editor, *Advances in Cryptology — CRYPTO '87*, volume 293 of *Lecture Notes in Computer Science*, pages 369–378, Berlin, Heidelberg, 1988. Springer. ISBN 978-3-540-18796-7. doi: 10.1007/3-540-48184-2_32.